

A Review of Artistic Interventions in the Development of Artificial Intelligence

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This review examines the evolving relationship between artists and Artificial Intelligence (AI), highlighting how artistic interventions shape AI's cultural and technical dimensions. In this study we observe artists taking varied roles in designing, curating, and critiquing datasets; adapting and exploring AI models and their outputs; and explaining and demystifying AI systems through art. Additionally, we discuss speculative futures of predictive technologies, considering how artistic practices reimagine AI's potential. The review identifies challenges, including balancing artistic and research agendas, and guiding AI's uncertainties. From this review we propose two strategic goals: continued work in mapping Creative AI territories and entanglements, and developing speculative AI devices through artistic practice.

Additional Key Words and Phrases: Creative AI, AI Art, AI Artists

1 INTRODUCTION

This survey explores the evolving roles of artists, and the contributions their works have had in the development of Artificial Intelligence (AI) as both a cultural and technical phenomenon. Artificial Intelligence (AI) systems are increasingly visible in the creative arts, where artists use these technologies not only as tools for the generation of media but as components of artworks, and subjects to be interrogated.

Over the past decade, the combination of Deep Neural Networks (DNN) and affordable computing hardware has facilitated a wave of technical achievements in computer vision, image generation, and audio synthesis, that has developed into the ever present “*calculative ambience*” [35] of the computational systems in our lives. Throughout this period artists have worked to understand the technical operation and cultural ideation of AI. In the last two years, the remarkable performance of AI Generated Content (AIGC) systems, such as ChatGPT and Stable Diffusion ¹, have dominated public conversations on Creative AI. The maturation of Foundation Models for image making [16], text generation, and audio synthesis [47], has led to the formation of new subgroups using these models [26]. New methods of interaction and exploration have been developed [140] alongside infrastructure for sharing and reusing AI models ².

Demonstrations of how artists can tactically engage with their entanglement with AI [138] go beyond the explanatory, and through experiential [61], and graspable methods [55], facilitating wider discussion on the potential of AI. Through the analysis and curation of datasets [106], design and reuse of Machine Learning (ML) models [3], and intentional misuse of commercial AI products [116], artists have blurred the lines of the roles of curator, maker, technician, researcher, hacker, and critic.

Concerns have been raised in relation to the role of AI in decision making, as well as the socio-technical issues in understanding AI - primarily under the term of Explainable AI [95]. As the cultural landscape becomes increasingly saturated with software outputs the changing role for artists in this environment has been questioned [127]. Past observation of artists' potential to raise “*critiques that may materialize in speculative alternatives to existing AI implementation and development approaches*” [138] has been realised in the development and integration of tools to address these concerns of consent ³.

¹<https://openai.com/index/chatgpt>
<https://stability.ai/news/stable-diffusion-announcement>

²<https://civitai.com>
<https://github.com/comfyanonymous/ComfyUI>

³<https://spawning.ai>

In a recent paper, researchers at the Creative AI Lab posed a number of questions aimed at initiating discussion on AI and cultural production with a focus on artists' involvement in "*back-end environments*" and developing "*new prototypes for technical processes*" [22]. These interests are echoed in the Serpentine Galleries' (UK) most recent annual strategic briefing - Future Art Ecosystems 4: Art X Public AI [123]- in which they discuss the entanglement of "*artists working across the AI tech stack*" and how this discussion has informed recent curatorial decisions and commissions within the institution. The interest of arts institutions in understanding AI is also evidenced by the provocation at the Liverpool Biennial in 2021 that "*the next Biennial should be curated by a machine*"⁴ in which the idea of an AI curated event could offer "*alien perspectives on conventional curatorial practices*"(ibid) in the hopes of generating new interpretations and connections between cultural works.

Addressing existing AI research, Tromble challenges the "*limited scope of art ideas addressed in AI research*"[128] and suggests artists join in making sense of the wide possibilities of what AI could be. These interests are different from a discussion on AI Art and artefacts, instead talk of the role both Artists and AI will play in the evolving culture.

In this survey, rather than only asking what AI can do for art, we also observe the types of contributions that artists have made in the development of AI systems, and in doing so, we examine how and why we should support this work.

"What art can do for AI is invite it into a realm with uncertainties and surprises, and ask it to play" [128]

2 RESEARCH QUESTION

The argument developed above led us towards the following question:

How have artists contributed to the development of artificial intelligence?

We will now move towards a systematic review of the literature to respond to this question.

3 SYSTEMATIC LITERATURE REVIEW

3.1 Scope and Selection Criteria

This survey brings together two collections of sources to explore both examples of academic research and artistic work. This approach is chosen to provide a comprehensive understanding of the contributions Artists have had to AI, and clarify the areas of overlap and mis-alignment between academic and non-academic research.

Overall this text combines the analysis of contributions emerging from two different corpora that have different intrinsic characteristics, so the selection method followed different approaches. For the systematic academic component we used the PRISMA method [97] with Scopus⁵ as the starting database. For the additional contextual component of artworks such a database does not exist, instead we followed a combination of analysis of well known resources⁶ and personal experience coming from 10 years of work in the field of computational art. We will further discuss the two methods in the following sections.

3.2 Systematically Selected Academic Research

The academic component of this survey is based on a collection of articles acquired through the Scopus database using the PRISMA method. Scopus was selected for its interdisciplinary coverage, which is necessary for the interdisciplinary

⁴<https://ai.biennial.com/>

⁵<https://scopus.com/>

⁶<https://creative-ai.org/>
<https://aiartists.org/>

subject matter of artists working with technical concerns. Scopus provides peer-reviewed research spanning relevant fields including Computer Science, Human-Computer Interaction (HCI), digital media studies, and art theory.

The articles were selected using a search term of keywords and boolean operators. Article abstracts were searched for the content “AI” AND “Artist” resulting in 412 documents. The addition of NOT “state of the art” was included to remove documents related to performance of AI models, which resulted in 400 documents. Further restriction to limit results to Articles and Conference Papers resulted in 309 documents. A manual reading of the abstracts was conducted to check the relevance of papers, removing articles that do not primarily address AI and artists. We also excluded non-English articles. This resulted in a final corpus of 86 documents.

Although we did not filter by publication date, these resources all fell within the past seven years, with all but one article published in the past five years, and 57% (49) of papers were published within the past eighteen months.

3.3 Non-Systematically Selected Artistic Database

The second collection of resources for this survey is made from a range of materials including research articles, artworks, interviews, art journal articles, and other reports or surveys from art institutions and galleries. These resources were collected over the past eighteen months by the author, and as such are a more opinionated sample deemed relevant from; personal and professional experience, acknowledgement of community and peer contributions, and inclusion of works previously recognised for their significance. This selection is important in contextualising the wider relevance of academic discourse and providing a focus for a wider audience.

3.4 Comparison and coding

The analysis begins with a coding and clustering process, where both collections of resources are reviewed for common themes, challenges, and methods. The non-Scopus selected materials were grouped into themes that emerged through the reading of the materials. For the systematically selected materials, we grouped the papers into themes that had been observed in the non-Scopus selected materials, and allowed new themes to be added where appropriate if they emerged from the reading of the texts.

These groupings resulted in the following themes.

3.4.1 *Datasets* : Scopus [60, 102, 113, 119, 137] Non-Scopus [30, 36, 46, 98, 100, 106, 111, 138?]

3.4.2 *Models* : Scopus [12, 18, 31, 33, 41, 50, 59, 64, 68, 77, 78, 81, 82, 84, 85, 87, 90, 91, 101, 103, 104, 109, 112, 120, 134, 135, 142] Non-Scopus [10, 13, 25, 26, 37, 45, 73, 75, 105, 107, 115, 116, 126, 141, 143]

3.4.3 *Explainability and Uncertainty* : Scopus [8, 17, 29, 63, 76, 124, 132] Non-Scopus [7, 14, 15, 21, 43, 48, 53–55, 57, 61, 61, 62, 89, 93, 95]

3.4.4 *Community and Concerns* : Scopus [4, 6, 11, 20, 24, 26, 32, 38, 39, 42, 44, 49, 58, 67, 69, 72, 79, 83, 92, 96, 108, 110, 114, 117, 125, 131, 133, 136] Non-Scopus [22, 34, 65, 71, 139]

3.4.5 *Future AI Systems* : Scopus [5, 9, 23, 27, 40, 51, 52, 56, 70, 74, 88, 99, 122, 130] Non-Scopus [1, 2, 19, 28, 86, 94, 128]

4 DEFINITIONS

To assist the reading of the review, here we offer some short definitions of terms used in this survey.

4.1 Artificial Intelligence

AI is a difficult term holding many interpretations that span technical and cultural domains. For our use in this essay we will attempt to offer two simple definitions, one technical and one cultural.

Technically, AI refers to systems designed to learn and perform complex decision-making tasks. These systems use algorithms and computational models to process data, recognise patterns, adapt to new inputs, and generate output responses.

Culturally, AI evokes ideas from science fiction, representing non-human or more-than-human consciousnesses. AI as a cultural or creative term challenges our understanding of our own intelligences, raising questions of agency, and how we relate to technology.

4.2 Creative AI / AI Art

The history of the term Creative AI is a term originating in the technical community in reference to AI systems designed for artistic purposes. Whereas AI art, or Machine Learning Art, is commonly used by artists working with these tools. Following the use of terms laid out by the Creative-AI Lab we use the term Creative AI where “*we are examining more than the artefacts ... focusing on Creative AI as a research field*”[22], and refer to AI Artist or AI art when talking about practitioners and their works.

4.3 Datasets

Datasets are integral to AI infrastructure, especially in Deep Learning (DL) architectures. These are used in training models to make informed predictions or decisions relating to input to a system. Their content can range from collections of images, texts, audio to other types of data. The rise of large foundation models, trained on extremely large datasets of many billions of samples, has also led to the increased use of fine-tuning datasets as evidenced by resources available on websites such as huggingface ⁷. These Fine Tuning datasets allow a user to bias a pre-trained generalised model towards specific use cases. Fine tuning can be applied through directly updating the weights of the model, or enhancing models through other processes such as LoRA [66].

4.4 Models

A Model refers to the architecture of the system that is designed to perform a specific task, such as prediction, generation, classification, or embedding. Models are made of components that process data in different ways leading to the output. These models are typically activated through a training process, learning from a dataset how to accomplish their task. The learning process involves iteratively adjusting the model’s internal parameters, often referred to as weights, to minimise error and optimise performance on the task. Over the past decade, Deep Neural Networks (DNN) have become the dominant type of model in AI, outperforming other techniques with larger datasets, more parameters, and greater computing power. The recognition of scale being critical to the performance of these systems has been referred to as “*The Bitter Lesson*” [121] by the reinforcement learning pioneer Richard Sutton. Bitter in the recognition that brute force methods and computational scale seem to repeatedly outperform more considered architectures for intelligent models (ref). DNNs take a variety of forms including: Convolutional Neural Networks (CNN) often used in computer vision tasks. Recurrent Neural Networks (RNN) that are designed for handling sequential data such as text processing. Generative Adversarial Networks (GAN) made from a discriminator and generator that compete against

⁷<https://huggingface.co>

each other in detecting and generating content that matches that of a dataset. Transformers which have spurred the recent wave of developments in AI capabilities with their self-attention mechanisms, seen in architecture such as Generative Pre-trained Transformers (GPT). Diffusion Models that iteratively refine noisy data to generate outputs, diffusion models have gained prominence in fields such as image generation.

4.5 Latent Space

Latent space refers to the intermediate, or hidden, layers within a model, particularly in neural networks and other complex algorithms. These layers represent the internal workings of the model, encoding information in a form that is often not immediately interpretable to humans but captures essential features of the input data. Guidance of models is often discussed in relation to latent space, where the manipulation of values in these layers can allow us to guide the system towards certain outputs.

4.6 Explainable AI

Explainable AI (XAI) is a field of study originating within computer science that aims to enhance transparency and interpretability of AI systems [95]. In addition to supporting the output of a system, XAI seeks to clarify *how* and *why* the system reaches a particular decision. This can involve a variety of techniques, including: analytical tools for monitoring latent variables and activations within neural networks, surrogate models that monitor the behaviours of more complex models, visualisation to translate complex internal processes into simplified forms understandable by non-experts, and debugging and auditing tools.

4.7 Entanglement

This term refers to the connection we have to the phenomena we investigate. Commonly understood through phrases such as *Life Imitates Art* or the media theorist John M. Cullin's statement "*We shape our tools and thereafter our tools shape us*"⁸. Entanglement has been related to ideas of cultural inscription where cultural narratives and values shape, and are shaped, by the work we do [80].

Throughout the rest of this paper, the use of these terms will follow these definitions.

5 REVIEW

In this section we will discuss the work that relates to each of the identified themes. For each theme we will first review the *Scopus selected* resources of academic research, and then we will look at the *non-Scopus selected* resources, before synthesising the findings.

5.1 Datasets

The design, development, and critique of datasets has been explored by artists and researchers. In this work, artists can be seen to be taking on mixed roles including curators of collections, or employees of data labelling companies. Here we see artists raising concerns over the content and consequence of certain datasets, and proposing tactics in closer readings of these datasets for better understanding the content and the processes of data collection.

5.1.1 Scopus. Within the research available in Scopus, the critique of the content of datasets, and the biases they represent, has been widely discussed. When we look specifically at datasets used for Creative AI models, Hakopian

⁸<https://www.themediumisthemassage.com/the-record/>

discusses how datasets derived from certain art histories have **reinforced ideas of aesthetic values** into new AI models [60]. Hakopian reflects on the AICAN (ref) art system, and discusses the system developers' claims that they have used a large training dataset of 80000 images containing "*the greatest works from art history... representing the Western canon over five centuries*" (ibid). Hakopian observes that there appears to be an uncritical filtering of the dataset, ignoring other possible creative outputs. Hakopian discusses how the art world has attempted to turn its attention to previously illegible or unseen art histories, whilst simultaneously, datasets are reinforcing the historically visible canon through their use in new image production. Questions of how datasets are funded, curated, and who is involved in this process, are used to investigate how we might assemble new datasets which may offer alternative creative visions.

Through interviews with thirty-three artists, Stark and Crawford look at datasets in **individual artists' practices** instead of large commercial projects [119]. Working with datasets and data collection processes, these artists focus attention on how datasets are developed and distributed. They note how this collection and assimilation of data tends to "*slip below the surface*" (ibid). The authors explain how using technologies in "*unexpected and often prescient ways*" (ibid) allows artists to reflect on data in their works, with one artist mentioning how "*I noticed that my own mistakes and faults and biases got implanted in the systems I was making*" (ibid).

Salvaggio also discusses their own artistic use of datasets through reflection on the process of working with their **personal data** [113]. By training a Generative Adversarial Network (GAN) on personal photographs, Salvaggio prepares a dataset not meant to be publicly shared, and in turn trains a model for private outputs. This work investigates how our relationship to data changes as we deal with personal memories becoming data. Shifting the viewpoint to a personal and emotional connection calls for different processes and this work invites us to rethink the importance of data collection methods.

Yu et al. approach the discussion on datasets not from an individual's perspective, but that of a **larger cultural group** [137]. They investigate the presence of the Korean emotion Han, to see if it can be carried through data and into a model. They prepare a dataset of melodies and lyrics from Korean government sources, alongside ballads, folk tales, and traditional music, then use this self assembled dataset to train a model that generates melodies and accompanying lyrics. Surveying 128 Korean people, alongside interviews with experts, they assess the presence of emotion in the model outputs. They find that many respondents do identify the emotion, but with caveats, for example one expert responding that the result "*depicts the vibes of the newly reinterpreted version of 'Han,' different from the way it is typically expressed traditionally*" (ibid).

The processes involved in **preparing datasets** was observed in the video installation artwork *The Cleaning of Emotional Data* by Elisa Giardina, a work discussed by Parisi [102]. Here the artist is drawing attention to the unseen processes that make up the development of AI systems. Whilst making this artwork, the artist worked for several "human-in-the-loop" companies who clean datasets for emotion detection, a process involving manual annotation of facial expressions and categorisation of emotion. Parisi uses this artwork to illustrate concerns of optics (in seeing as a primary way to understand the world), surrogacy (where bodies teach and correct the AI), and suggests that allowing for negative-optics in the assembly of datasets could allow AI ways of understanding not bound to human perception.

5.1.2 Non-Scopus. Looking at the *non-Scopus selected* resources we see similar concerns. The well known artwork *ImageNet Roulette*[36] highlighted problems with classification models built upon the ImageNet dataset by letting the dataset "*speak for itself*" (ibid). Zeillinger discusses this work as an example which shows "*alternatives to existing AI implementation and development approaches*" [138], where the appropriation of this dataset outside of its intended field initiated a conversation on the utility of the resource. The results of the work plainly showed a visible mislabelling of

images, that was not just a mistake in the labelling but a result of choices made in the design of the labelling process. By using the dataset in *ImageNet Roulette*, they were able to publicly demonstrate the problems with building systems uncritically.

In their article *On Lacework* [106], Everest Pipkin recounts the experience of watching one million three second videos of the MIT Moments In Time video dataset [98]. They reflect on being “*probably the first person to watch*” all of this “*unremarkable*” dataset, and how it feels coming into contact with cultural material “*meant for machine comprehension, not human*”. Curation of exhibitions, or catalogues, or museum archives, is a role many artists have, but with large scale datasets this curation is segmented and distributed to workers around the world, who each see small segments and make judgements about the inclusion of samples. Pipkin most likely still remains the only person to have seen this dataset, the only person to experience the difficult process of viewing moments of “*extreme emotion and personal vulnerability*” (ibid) embedded in the collection. Through their writing they ponder “*if this is how a sorting algorithm feels*” (ibid) as it scrapes through the material, and say how they came to a new understanding where their body “*learns the rules and edges of the dataset*” (ibid).

Before the use of pre-trained foundation models became popular, many artists working with AI typically assembled their own **custom datasets** for their projects. Anna Ridler collected 10000 images of tulips for the work *Mosaic Virus* [111], Jake Elwes’ *Zizi Project* proposed a process of “*queering the dataset*”, where they retrain models with 1000 images of drag and gender fluid faces added to the original training datasets [46]. Mimi Onuoha’s ongoing project *The Library of Missing Datasets* [100], again highlights the “*blank spots in spaces that are otherwise data-saturated*” (ibid), showing what we pay attention to and what we ignore.

As well as producing datasets, artists have designed new data collection pipelines. Adam Chodzko developed *The Dreamshare Seer* [30] a **participatory project** with residents of the Isle of Sheppey, building a database of dream reports which lays a foundation from which future work can develop. Here the artist is involved with the community in collecting and curating the content of the dataset over time. Similarly, Caroline Sinderson’s project *Feminist Data Set* [118] includes an open source toolkit with which others can interrogate the entire dataset process, from collection and labelling of data, to training and using models.

5.1.3 Summary on datasets. What we see in both the *Scopus* and *non-Scopus* resources are examples of artists making sense of computational systems through taking on roles as curators [46, 111], critics [106], developers [118], labourers [102], and community organisers [30]. From large scale datasets used in commercial products, to small, hand-crafted datasets; we witness the scope of the works, from the public, to the personal. The investigation and intervention in the content and collection of datasets helps us understand the processes involved, the problems emerging, and possible paths for working with datasets.

5.2 Models

In this section we look at the roles of artists in developing and using AI models. This includes the design and implementation of models, adaptation of existing models to new uses, inference and sampling techniques, and research on the utility and application of Creative AI models.

5.2.1 Scopus. Guo et al discussed the use of AIGC models in bringing together **traditional artistic practices** and technology [59]. Through case studies and interviews with participants they demonstrated how the use of commercial

AI tools ⁹ could assist rural grassroots art communities working with their local cultural heritage. Li and Liu also explored the use of style-transfer models as educational tools, to assist artists in mastering traditional Chinese painting techniques, and discuss the use of this technology in allowing more people to “*contact, learn and create Chinese painting*” [81]. Coming into contact with traditional artworks through the use of technology was also explored in the interactive artwork developed by Zhou, where visitors to the exhibition used their own body movement through the exhibition space to provide input to a GAN which generated Shanshui ink brush paintings [142].

The use of AI models for keyword extraction and topic expansion in a mind-map-painting application was demonstrated by Liu et al [84], who combined Shanshui painting with diagrammatic outputs. Contextualising artworks using model recommendations has also been used in the development of tools for art historians to build narratives around collections [12].

Regarding artistic **control** of models, Wolfe et al combined the traditional crafts of printmaking and book making with image generation models [134]. In a classroom study they noticed how participants were “*initially frustrated by a lack of control*” (ibid) but how they came to enjoy finding “*ways to react to those results*” (ibid). Lyu surveyed artists and non-artists (N=42) to assess the outputs of image generation models, also noting “*artists were not satisfied with the control effect of AI, and they even felt a little out of control*” [90]. Li reflected on their own practice of using the image generation model Disco Diffusion for generating AI Paintings in a variety of styles over the course of three months [82]. Then also surveyed experts and non-experts opinions to assess these images, finding that it “*performs poorly on Chinese ink painting*” (ibid) due to misuse of colour and “*chaotic*” composition. Wu et al also mention artistic problems with stability and similarity of output of GAN models [135].

Looking at the **failures of models**, Ou developed a prototype system for AI aided 3D model processing [101]. They discussed findings from two artists who used this tool daily in their work for three months, and further investigated these findings in a lab study (N=20). They observed that failures in this process often originated with human error, which was then incorrectly addressed by the models preferences. The preferences of models can introduce uncertainty and occur in both inference parameters and original training dataset.

Model failure was also discussed in Krol et al study of GPT-3’s ability to communicate musical decisions [77]. By inputting textual representations of music they found GPT-3 could demonstrate some understanding of musical structure, but suffered from a “*lack of consistency*” (ibid) and was unreliable. Intentionally inducing failures was also seen in Ferner’s work BoschBot [50], in which small segments of Hieronymus Bosch paintings were fed into an image captioning model, an intentional misuse of the technology leading to “*curious interpretations*” (ibid) that helps communicate the internal behaviour of the model. Parslow analysed the work of Jake Elwes, and how Elwes’ project did not “*take itself too seriously*” [103] which allowed it a playful but sensitive interaction with the failure of machine vision. Pereira reversed this tactic of using artwork to give us new ways of seeing AI models, and showed how using commercial AI tools to look at artworks showing how “*AIs can work, through their glitches, to level and reimagine artworks, giving us a fresh set of eyes for understanding art*” [104].

The **failure of models** was also looked at from the perspective of “art hacking”. Kronman discussed how artists have hacked machine vision models. This practice is more performative, creating “*dispute rather than providing solutions*” [78]. Rodier et al. demonstrated a process of working with artists, engineers, ethicists, academics, and makers as a way of examining AI systems [112]. Bozzi also focuses on machine vision models through analysis of a number of artworks

⁹<https://www.midjourney.com/>
<https://app.runwayml.com/>

used as case studies [18]. Similarly, Ramakrishnan uses artworks to discuss the operation of machine listening models [109].

Other works are less concerned with interaction or accuracy of models, and instead look at our **entanglement** with the content generated by the models. Marynowsky's video artwork *From our deceased bodies flowers will grow, we are in them and that is eternity* is made from images generated using the Artbreeder GAN model [91]. The artist created images of evolving flowers, and through this process human faces would appear, showing how a human tendency to see faces in the world around us can also feed into the models we develop. Cole and Grierson looked at the human quality of desire in AI models outputs through the development of the audio-visual installation *Kiss/Crash* [33]. Henrich et al discuss the structure of neural networks in relation to their own video work through a conversation with GPT-4, looking at the romantic and poetic potential of AI models in an attempt to understand "*what distinguishes you from a text generator*" [64].

Interaction with AI models takes many forms. These forms are seeking to find ways to interpret and guide model outputs by manipulating inputs and layers of neural networks. The process of manipulating model output through latent space navigation has also been compared to the artmaking process [41]. The process of making with AI was also documented by Huang et al [68], who used AI in phases to generate a video of the dream of Mulan, with artists guiding inference towards concrete components of a larger work. As large AIGC models have become popular, developments with text and multimodal prompts have assisted in navigating the wide range of possible outputs [85]. Negative prompts have also emerged as a mode of interaction with text-to-image models. Negative prompting is guiding the model through what you do not want to generate. Through the lens of theology, Lucas has discussed negative prompting in relation to the ineffable qualities that are of interest to artist-researchers [87]. Chung discusses the potential of AI Creativity Support Tools (CST) that provide "*idea-wise support*" [31]. The use of text models as **conversational tools** to interact with artworks as they develop, has been explored with *ORIBA* [120], a tool that allows illustrators to converse with their characters. In this work we see creations start to become co-creative partners.

5.2.2 Non-Scopus. Within our second collection of resources we see how artists' use and investigation of models has changed over the past few years. Before 2021 and the wave of new large foundation models, most of the well known artists working with AI were focused on the use of **image generation** GAN models. Helena Sarin used GAN models in **Leaves of Manifold** in a practice she calls Neural Bricolage [115]. For this work she trained her own model on a modified CycleGan architecture [143]. Gene Kogan's *WikiArt GAN* [75] used the Progressively Grown GAN [73] trained on 80000 images, with the potential images explored through interpolating through the latent space. Mario Klingemann's *Neural Glitch* series investigated the results of altering, deleting, and exchanging trained weights in GAN models to introduce glitches into the images [107]. Bawa-Cavia performed similar data injections into a Long Short Term Memory (LSTM) Network, exploring how this disruption could help "*reveal formal aspects of the network's architecture*" [13]. In their work *The Latent Future*, artist and researcher Nao Tokui looked at the generation of "*near endless combinations of texts*" [126] by language models, creating news feeds that are fictional yet exist as potential events in the model. Despite all of these examples we still lack technical details on how the artists undertook their work, reiterating a point made by Cetinic and She, who, in their survey on the use of AI for art analysis and for generating artworks, note that it is often unclear how artists are using models "*as many artists do not reveal all details of their creative process*" [25].

Erica Scourti's *Exit Script* [116] used a **commercially available** emotion detection model to rearrange personal diary videos by emotional content. The service was originally designed for monitoring of staff working in call centres,

and through the intentional misuse of the service the artist reveals the lack of transparency in this system, as well as exploring the potential of models to offer “*different kinds of biographical writing and bodily inscription*” (ibid).

The **latent potential** of models has been explored by Pfefferkorn and Sunde who discuss the emergence of a character “haunting” the latent space and repeatedly appearing in images [105]. They relate the wider discussion of this character to the audience’s fears of technology, and suggest these evocative findings can be useful in making technological processes visible. Another often observed quality that appears in text-to-image models is in the difficulty they have in generating images of human hands. Balbontin discusses how generating images of hands raises questions of errors and incoherence, suggesting the seemingly incorrect AI image can be reinterpreted as opening up the idea of what is possible in the real world [10]. Weidi Zhang’s interactive virtual reality installation *LAVIN*, allowed the audience to “*visually explore one understanding*” [141] of AI, where the camera feed of the real world is classified into fifty objects and visualised in a projection, offering an illustration of the operation of large scale classification models. Across these works we see how the qualities of models that may be seen as mistakes or misreadings can be repositioned to inspire rethinking our own understandings of the world.

Elliot and Grierson discuss the recent trends in Creative AI as distinct from earlier GAN AI Art [45]. Changes in the architecture of models have modified the ways models are used. With the availability of cloud GPUs artist-developers began sharing Jupyter Notebooks for running models, for example repositories such as ¹⁰ and the well known VQGAN+CLIP notebooks [37]. This popularity of text-to-image models led to the emergence of Prompt Artists, who design and share textual language templates [26]. Developments in models can result in new interactions and unexpected new subcultures.

5.2.3 Summary on Models. In this section we have seen how artists have explored AI models. We have seen studies that have highlighted issues of control and failure of models [134], and artworks that embrace this failure or offer new ways of interacting with models to address control [116]. We have also seen works that examine the content of image generation models and how this can be read not only as a failure but as a quality that helps us reimagine our own perception [10].

Other research has shown how models can be used in cultural heritage projects to support artists working with traditional techniques [59]. As this technology has matured practices have changed, from individuals training custom GAN image models, to communities prompting large scale foundation models [26].

5.3 Explainability and Uncertainty

The section discusses the tensions between the function and utility of art in communicating technological processes and the difficult and unexplainable qualities of these processes that appeal to many artists. Here a new role to be played has emerged, but with it a potential dissatisfaction.

5.3.1 Scopus. Botz-Bornstein discusses the importance of ‘randomness’ in both human and AI art, contrasting AI’s preference for certainty with artists’ embrace of uncertainty [17]. This discussion clarifies how randomness is not the same as uncertainty, but often used as a substitute in computational art, and raises questions on the consequence of this for artists who thrive in speculative and uncertain environments.

The difference in intentionality between humans and AI was discussed by Terzidis [124], where they position speculation as used by artists as going against what AI wants - certainty. Ashby explores a participatory speculative

¹⁰<https://github.com/lucidrains/deep-daze>
<https://github.com/lucidrains/big-sleep>

process to envision possible AI futures [8]. Chesher pushes this uncertainty further, talking about AI in relation to magic [29], whilst Kratky draws historical lines to the approach of the Surrealists, noting an attraction towards AI's capacity for surprising juxtaposition of ideas [76].

Hemment et al have led developments in AI literacy through the use of artworks, nurturing an idea of **Experiential AI** as a widening of Explainable AI where they examine the utility of "*interpretation over explanation*" [63]. Walker et al. discuss a similar process of developing eighteen AI artworks with the specific aim of countering misinformation [132]. They spoke of "*hard-to-refute strategies*" (ibid) bringing narrative and emotion to conversations on deepfakes and data literacy, promoting this form of research and saying AI art tools are deserving of "*further investigation as to their effectiveness*" (ibid) in public communications.

5.3.2 Non-Scopus. Other research that did not appear in the Scopus search has also looked at AI artists and *uncertainty*, particularly with relation to predictive qualities of AI models as a form of divination. Browne discusses the operation of neural networks through an interactive computational ouija board [21]. Benabdallah demonstrated a combination of oracular traditions and machine learning with a predictive home device [14]. Dukes discussed the self-fulfilling prophecy of AI and how this will "*profoundly alter our ideas about magic*" [43]. Gilchrist spoke of a "*dearth of language*" around AI that has led to this mystification, and grounded his discussion of AI in the language of remix culture [57].

Explainable AI originated within the field of AI as researchers looked for ways to communicate the inner workings of AI systems. Communicating not only what decision was made, but why the system came to this decision. Miller explains the idea of Explainable AI "*that by building more transparent, interpretable, or explainable systems, users will be better equipped to understand and therefore trust the intelligent agents*" [95]. The overlap with the concerns of many artworks previously mentioned seem present within this definition. However, this idea of bringing Explainable AI into the arts has been **criticised** by Arora and Sarkar, who reject the using artworks in explaining AI systems as "*a symptom of insecurity of the anthropocentric notion of art and a nostalgic desire to return to outmoded notions of authorship and human agency*" [7]. They discuss how explanations can fall into a trap of becoming ornament, where the artwork as decoration feeds into mystification, a danger of critique that aligns with what Berger described as "*explaining away what is otherwise evident*" [15]. Resisting the simplification of our lives and the world we experience in an attempt to make it legible to a system can be seen in the work of Lustig and Rosner who discuss how the aims of Explainable AI (accountable and transparent systems) must also reflect the parts of ourselves difficult or impossible to be explained [89]. Lustig and Rosner explored this joining of explainability and uncertainty through the design of a Tarot deck using AI image generators, and through this work they reflect on the difficulty in highlighting the importance of the unknowable and ineffable within computational systems.

Expanding on Hemment et al. paper in the Scopus texts section, the idea of **Experiential AI** as a development of Explainable AI has been further discussed as a "*new research agenda in which artists and scientists come together to dispel the mystery of algorithms and make their mechanisms vividly apparent*" [61]. Working with artists previously mentioned in this paper (Anna Ridler, Caronline Sindors, Jake Elwes), these researchers have developed a framework for conducting this research [62] and demonstrated its use in designing intelligent interactive systems that can make AI systems understandable through "*informative and compelling experiences*" [61]. Falcone and Castelfranchi's work on autonomous agents can be seen as supporting this idea, in showing how trust is developed through direct experiences [48].

Similarly, within the field of tangible and embodied interaction, Ghajargar et al have promoted the idea of Graspable AI with similar aims and claims to offering a new perspective on understanding and explaining AI [54]. Ghajargar and

Bardzell developed ‘concept cards’ to assist in the design of Graspable AI [53] and demonstrated the use of this concept with Physical Artefacts [55]. McGrath discussed the design of AI systems through the craft of making them, bringing understanding of AI through material thinking that could “*leverage the potential of sensory, tactile experiences to spur reimaginings of AI*” [93].

5.3.3 Summary on Explainability and Uncertainty. The recent research interests in the usefulness of artists in explaining AI systems through artworks has been promoted as providing a better way of communicating the operation of AI, but also criticised as removing some of the magic in the process. Some artists have embraced the unknowable and uncertain operations of AI as a quality that brings their work into conversation with the greater unknowns, while others have looked more into the usefulness of art to bring clarity to complex systems through engagement with a wide range of senses. The recent development in framing the discussion of AI artworks within *Experiential AI* or *Graspable AI* could help in finding a fruitful overlap between the utility of artworks in communicating to a wide range of senses, as well as encouraging the connection to uncertainty and unknowability of AI.

5.4 Community and Concerns

Examining the impact of current AI tools on artists’ practices alongside the impact on their communities is important in understanding how roles are changing or will change in the future. In this section we review concerns that have been raised alongside the changes in practices and community reception of AI art tools.

5.4.1 Scopus. Through interviews with artists and scholars working with Indian art forms, Divikaran explored the **cultural perspective** embedded in human art, and proposed rethinking the discussion on AI ethics as a participatory process rather than working to fixed frameworks [42]. Baradaran discussed how “*we must begin to create tools*” [11] that offer different stories and understandings to the ones otherwise crafted for us so that we can reimagine the role AI takes in society. Understanding the tools that have been made, and their influence over artists was discussed by Issak and Varshney[69]. In a rejection of the entanglement of society and technology, Wojtkiewicz claimed that the social objections should be separated from criticisms of technology [133]. While Elam discussed race theory in relation to AI development saying “*the grandest technological challenge may in fact be cultural: the way AI is shaping the human experience*” [44].

Regarding **environmental concerns**, Deng et al. prototyped a design artefact in a study with AI artists with which an artist could receive information on the environmental impact of their proposed ideas [39]. They found this to be an accessible way to engage AI artists with sustainability. Similarly, Leibowicz et al developed a *Responsible AI Art Field Guide* for artists and designers to aid in adapting their practices whilst keeping these concerns in mind [79].

A number of studies have looked at **public perception** of AI generated artworks. Demmer found participants (n=48) reported stronger emotions with works made by humans [38], and Ragot showed a strong preference for human generated work over AI (n=555) [108]. Brewer found that exposure to certain media messages strongly influence people’s opinions on AI art (n=1035). Arkhipova and Viidalepp looked at the reception of AI generated videos of faces [6]. Mazzola et al surveyed visitors to an exhibition of AI generated images finding “*those who are more curious about the technological innovations are, at the same time, more fascinated and more frightened by the new AI tools*” [92].

Looking at the **impact of AI** on artists, Browne discusses the distinction between ‘artists working with AI’ and ‘AI Artists’, showing the impact of markets and AI issues in the elevation of some artworks over others, and claiming that AI has reinforced conservative forms and aesthetics [20]. In their systematic review of the impact of AI on the world of art, Then et al. observed “*significant impact*” in AI enabling immersive works of art, as well as increasing

audience interaction and participation [125]. Rani et al discussed a revolution in creative work that provided potential benefits in exploring technology [110], while Jiang et al. raised concerns over economic loss and reputational damage to artists whose work has been used as training data in large public models [72]. Vimpari et al described an “*adapt or die*” perception of AI present amongst some artists [131], with Allred associated these feelings towards AI resulting in communities rejecting these tools [4]. Although we have covered many artists choosing to work with AI in this survey, there are many more that are encountering tools that are difficult to understand and “*usually involves working with skilled technologists*” (ref explorers of unknown planets).

In attempting to **address some of these concerns**, Liang et al developed a method to prevent imitation of digital images, through perturbation of images not visible to the human eye but damaging to a models learning process [83]. Shan et al presented a similar image perturbation tool and surveyed 1000 artists to assess the usability of such tools finding it to be highly successful in protecting images [117].

Chung et al. interviewed fourteen artists to understand the relationships between artists and their support networks; collaborators, subcontractors, etc. Whilst mapping out what support is provided by the network they discussed how AI tools may replace and automate parts of this network, and raised how understanding the existing networks, the roles of participants and the ways they support artists, is critical in designing interactions with new tools [32].

A number of papers looked at the **practices of artists** using AI and the **communities** around them. Through a series of interviews with well known artists working with AI, Caramiaux and Fdili Alaoui studied the changes in artists practices, where the practice involves crafting algorithms and data as material [24]. A similar study was conducted by Gruchy [58]. Milovidov described how the restriction on content generation and model access of large companies has led to communities organising around their own algorithms [96]. Feffer et al reported on the culture inside the deepfake music communities in AI Hub Discord server ¹¹ and uberduck website ¹² dedicated to training models on collections of music, noticing training on artists from a wide range of backgrounds and levels of popularity, and calling for more research on these communities [49]. Huang et al discussed submissions to the AI Song Contest ¹³, finding more work needed to be done on making models “*decomposable, steerable, interpretable, and adaptive*” [67]. Sanchez surveyed practitioners on their use of prompts finding the use of artist names to be the main component of prompts [114]. Chang et al discussed the motivations of Prompt Artists, suggesting further work on tools for aiding novelty in outputs and validating originality [26]. The development of an autonomous AI organisation generating and selling artworks on the blockchain was explored by Yadlapalli et al [136].

5.4.2 Non-Scopus. Other studies outside of the Scopus selection relating to **public preference** in AI and human artworks have shown different results. Cooke et al found people (N=1276) struggled to discern between synthetic and authentic images [34], findings supported by Park et al survey (N=240) [71]. Hong and Curran observed (N=288) that knowing the author of the work to be AI or human also affected the attitudes participants had towards artworks [65].

The ability of AI to “*amplify*” [139] creativity has been described as a larger **disruptive force** for creative labour than automation. Tools to counter this disruption by addressing the ability of systems to copy digital images have also been developed and adopted by artists. The organisation Spawning AI has provided a number of resources including

¹¹<https://discord.gg/aihub>

¹²<https://www.uberduck.ai/>

¹³<https://www.aisongcontest.com/>

opt-out lists ¹⁴, standards for “bot” permissions through ‘ai.txt’, and distributed anti-crawler systems that block “bots” from accessing artists websites ¹⁵.

Organisations such as StabilityAI ¹⁶ have engaged with Spawning AI’s resources to remove non-consenting artists from their training datasets. The role of artists working within businesses can be traced back to examples such as programmes like the Artists Placement Group in the UK in the 1960s. The idea of artists working with and **within the organisations** developing AI is an open issue, with the Creative AI Lab asking “*Should we strive for artistic research to have systemic impact?*” [22].

5.4.3 Summary on Community and Concerns. As AI tools have become popular the economic changes for artists has caused some communities to reject these technologies [4], as others adapt their practices to embrace them [24]. The audience perception of AI on images is unclear with conflicting studies [34, 108] similar questions remain to be studied about other AI generated media. As such the long term impact on the employment of artists remains uncertain. Researchers have developed tools to defend artists’ images from becoming part of large training datasets, and these have been successfully integrated into leading Creative AI organisations’ workflow.

5.5 Future Systems

Where artworks presented in previous themes tend to engage with past and present real world forms of AI, this theme looks at how artists have reimagined the potential of AI. These works looking at co-creative partnerships and living systems designs are far more speculative, and reimagine the potential roles that artists could have in interacting with other forms of intelligence.

5.6 Scopus

A number of research papers have speculated on the development of AI and what AI might look like in the **far future**. Nikolic proposed the future of AI taking two possible paths; replicating and mimicking human mental processes, or finding a new idea of independent machine creativity [99]. These paths can be seen in the work by Kim discussing their process initially using AI as a tool then bringing inspiration from the AI into their work. Reflecting on this process they imagine a future where “*AI will no longer be recognized as a new tool but as a new ‘partner’*” [74]. Amerika’s *Artificial Creative Intelligence* also acts as a co-creative artistic partner in writing poetry [5]. The ethical considerations of such authorship of future sentient machines has been speculated on by Galanter [52]. Lundman and Nordström interviewed twenty-six artists using AI in their work, finding novel co-creative practices emerging [88].

Audry further opens up the possibilities of AI by going **beyond computational thinking** into the intelligence of living systems [9]. They discuss how the learning process is durational, a performance, and how artists working with intelligent, learning models need to “*provide appropriate contexts to support the natural unfolding of these new behavioral forms*” (ibid) so that audiences can “*get to know them*” (ibid). Others have also seen the AI as a co-creator to “get to know”. Tang looked at the mimicking of artists’ style by a robotic calligraphy model [122], working with robot-human AI systems was also discussed by Chen [27]. In conversation with ChatGPT, Giannachi discusses AI-human entanglement and how it has been explored by a number of artists [56]. Jansen and Sklar’s study of drawing practitioners looked at humans drawing alone and together, humans drawing with AI, and AI drawing alone to better understand the working environment for drawing for designing future systems for collaboration [70].

¹⁴<https://haveibeen trained.com/>

¹⁵<https://kudurru.ai/>

¹⁶<https://stability.ai/>

Forbes discussed how moving towards **multisensory systems** and multimodal models is an opportunity for AI systems to move from mimicry into critical inquiry [51]. Vierira et al. [130] look at the emergence of an interdisciplinary AI field in the internet of multisensory, multimedia, and musical things. Moving towards embodiment and multisensory intelligence is also discussed by Denson [40]. How these more-than-human intelligences would develop is uncertain. Cakl discusses learning from the Mould cultures in Ivor Diosi’s artworks as a way of imagining a less-human (or more-than-human) AI [23].

5.6.1 Non-Scopus. McLuhan spoke of art as “a Distant Early Warning system that can always be relied on to tell the old culture what is beginning to happen to it” [94]. Many of the engagements of artists working with AI talk of immediate and present concerns. Opening up possible futures allows us to discuss the greater potentials within recent developments. In his final book, Lovelock extended ideas from his influential Gaia theory, into that of a *Novacene* [86], a proposed geological era dominated by intelligent machines, that he describes as already underway. The idea of new forms of thinking, new ways of being, and alternative ideas of intelligence has been discussed by Bridle [19]. Offering an ecological framing of AI, one that can bring the flexibility and resilience of living things into computer systems and infrastructure, can allow us to reimagine AIs potential. Ongoing research into new forms of computational hardware based on living systems principles is being explored by Ackley [2].

Ian Cheng’s *Emissaries* series explored cognitive evolution and the “ecological conditions that shape it” [28] with simulated environments using AI to develop a story of infinite duration. The collective Orphan Drift have explored machine and human vision for 30 years, most recently imagining AI as a “distributed, many-minded consciousness” [1] in an octopus-like form. Pushing away from operations and motives of commercial AI projects gives these works space to “*feelthink and perform our knowledge that the world is more than human, greeting the other ... with a radical act of love*” [128].

5.6.2 Summary on Future Systems. The repositioning of AI from a tool to a co-creative collaborative partner, or even further to alien consciousness we can learn to communicate with, shifts the role of artists significantly. Although ideas of future AI systems are being explored, the development of tools and tactics that can support these practices is scarce beyond artists developing their own resources. This may be partly due to lack of commercial application as much as the fluidity and uncertainty in the ideas being explored.

6 FINDINGS

In Figure 1 we illustrate the previous section of the review, by embedding the paragraphs of this section to convert texts into lists of numbers representing the content, clustering with K-nearest neighbour and cluster optimisation techniques, then projecting the high dimensional embeddings into two dimensions using T-SNE [129]. Keyword summaries for each cluster are generated using GPT-4, and contours are added to help visualise the distribution of points. This diagram helps aid our overview of the field.

In this review, we have explored how artists engage with and influence the evolution of Artificial Intelligence. We will now discuss the findings, identify existing challenges, and highlight potential areas for future research.

6.1 Datasets

6.1.1 Findings. The first theme we discussed was *Datasets*, where we reviewed research and artworks investigating both the content of datasets, and methods of data collection. Assuming varied roles including data curators and data labellers, artists demonstrate how unconventional approaches to data analysis can uncover insights into the cultural

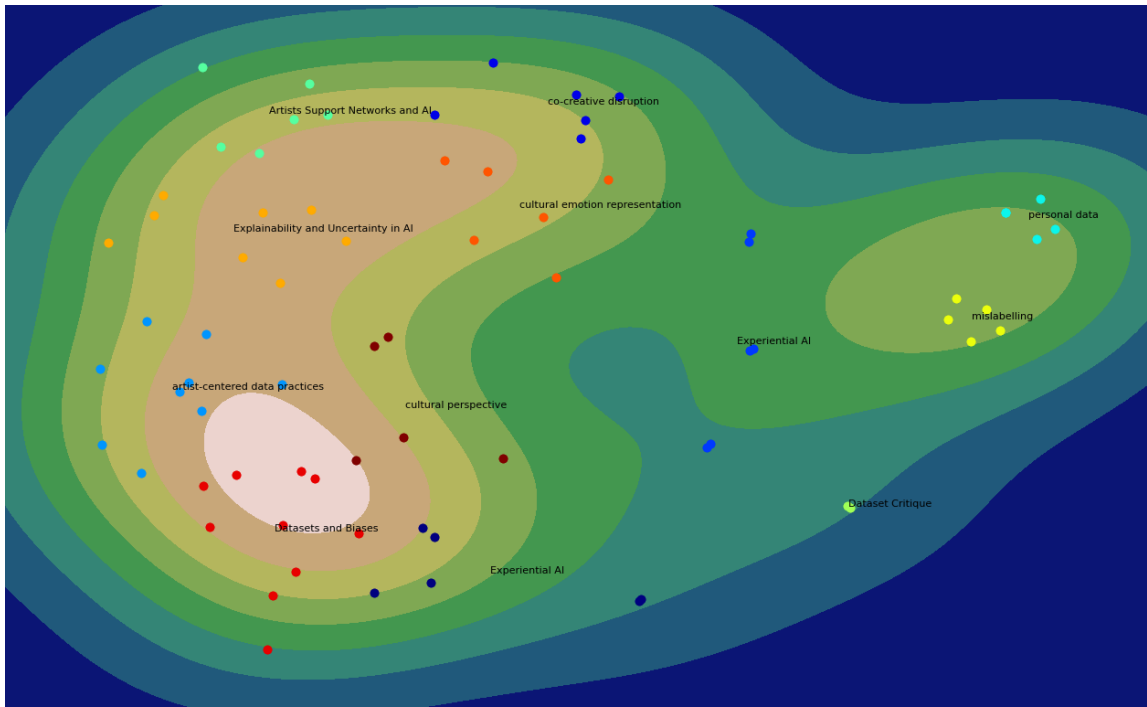


Fig. 1. Clustering and contouring of the review section of this survey.

ideas of AI systems. Through the works reviewed under this theme, we observe how artists are engaging with datasets and the opinionated content contained within those datasets. Despite technical changes in dataset collection, many artists are still assembling and curating their own datasets for personal artistic use. Others have proposed processes for participatory assembly of datasets to include other data beyond their personal selection.

6.1.2 Existing Challenges. A practice that has persisted from earlier artistic works to more recent artistic works is the assembly of datasets by those artists for their own use. There is a clear absence of technical detail regarding the tools used by these artists for dataset curation. Since the start of the period of these publications, five years ago (2019), data collection methods and tools have evolved. Earlier artistic works discussed popular datasets that are no longer available. It is unclear if the methods of dataset interrogation applied to earlier, smaller, datasets still apply to the new systems.

6.1.3 Areas for Future Research. In this section we will pose questions arising from our analysis of the literature to identify areas of potential future research.

How can artistic investigation into datasets of unconventional media provide new insights into AI systems?

When we consider artistic intervention into datasets of unconventional media we focus upon the re-application of tactics and tools developed in past works. We note how most artistic interventions in datasets are related to traditional creative media; namely image, audio, and text. This raises a question about the use of artistic investigation into datasets of other types of media less commonly related to creative outputs. For example, stock market and financial time series data, which would not normally appear in creative works but is often studied in relation to the application of AI.

What methods and tools can assist artists in examining the content and assembly of synthetic, machine-generated datasets? How can these support practices related to self-assembled datasets at a scale suitable for modern Creative AI systems?

With the recent shift towards large scale foundation models, and their newer data processing pipelines, the assembly and usage of datasets has changed. The scale of these datasets has gone beyond that of an individual's capacity to engage and curate. Research into the reuse of previous methods, and development of new tools can help examine changes in technology.

6.2 Models

6.2.1 Findings. Our second theme focused on *AI Models*. These are the functional components that create output in the form of generated media, classifications, predictions, or other representations such as embeddings. Researches have shown how AI models can support cultural heritage projects as well as traditional creative work.

We saw how in the recent past many artists have developed and trained models for personal creative use. These practices have changed with the popularity of newer pre-trained large models, and new modes of interaction have evolved with new communities engaged in this development.

6.2.2 Existing Challenges. Following the analysis of the literature it is evident that researchers have engaged in studies with a wide range of artists' communities. These studies report that artists are expressing frustration with models when working in traditional practices [90]. Despite this we can see new subgroups developing within active communities, who are using AI models in their own practices to suit the needs of their work. There is a lack of alignment between researchers examining the potential use of Creative AI models, and the practical use of those models by other artists.

New large models are difficult for artists to train themselves, consequently more dependance is placed upon AI organisations to support the maintenance of infrastructure and development of these models. Alongside this increased dependency on third parties, interfaces have become more standardised. It is unclear how these issues have affected the artistic investigation into the architecture of models.

6.2.3 Areas for Future Research. **How does the sense of control, or lack thereof, affect the interaction between users and AI models?**

Continued work into understanding control in the interaction with AI models, particularly for creative outputs, is necessary. This involves not only finding ways for individuals to guide models effectively, but also recognising whether the reported frustration arises from the resulting output or from the artist's desire to make a controlling decision. It is important to understand ideas of control alongside the use of uncertainty to connect artwork with the unknown, and bring this understanding into the design of AI models.

What are the socio-technical dynamics of new AI communities, and how do these influence the evolution of interaction techniques and architectural designs?

Observing the emergence of new communities around new architectures and interaction methods could provide valuable insights into the wider adoption of AI. Research into emerging techniques and how they develop could be fruitful for further study. Examining how these new communities support themselves through tool-making, model sharing, discussion, and critique can help inform the design of future AI systems.

6.3 Explainability and Uncertainty

6.3.1 Findings. Researchers have discussed how artists have embraced ideas of uncertainty, or even magical thinking, in their interactions with AI. This connection to the unknown and unknowable aspects of non-human processes is particularly powerful in opening discussion on what AI is as a technical and cultural idea, in that it relates to other cultural practices such as divination and the histories of prediction. Parallel research has found a functional approach to integrating art practice within Explainable AI, where artworks play a role in bringing new senses to explainable AI.

6.3.2 Existing Challenges. There are conflicts between the aims of artistic connection to the unknowable, and the research aims of art as a useful way of expanding ideas of knowledge. There is a possible union of these aims, as they can both look towards a holistic understanding of AI. There are visible difficulties in managing where it is appropriate to allow space for artistic indeterminacy and where it is necessary to seek practical clarification. In this emerging field it is necessary to explore methods for artistic research that do not compromise the development of the artwork for the aims of research agendas.

6.3.3 Areas for Future Research. **What is the role of artistic research in framing the divinatory and uncertain potentials of AI systems, and how can this framing contribute to the development of explainable AI (XAI)? How do artworks that explore AI systems differ from those that contribute to understanding them, and how can this distinction shape the role of artists in the field of explainable AI?**

In investigating the potentials of art in explainable AI systems, clarity is needed to highlight the difference between artworks developed to understand AI systems, and artworks that we can read as contributing to an understanding. Clarity on the roles of artists here can help direct in understanding if explainability is a quality to be pursued by artists as makers, or as a curatorial framing of works.

6.4 Community and Concerns

6.4.1 Findings. Within the *Community and Concerns* theme we looked at the impact Creative AI was having on working artists and the response from art communities to the presence of new tools and systems. Looking at the public perception of AI generated media we see conflicting studies, some studies showing that audiences do not recognise a difference between human created and AI generated media, whereas other studies show the opposite observation.

Uncertainty around concerns such as audience perception, as well as economic or sustainability concerns, has led to resistance in some art communities to adopting Creative AI tools. At the same time, new communities have emerged through the introduction of new models and modes of interaction, such as prompt artists and deepfake music communities.

Artists have led the development of tools, both in supporting the use of AI models, and in resisting the impact of large organisations' models on their practices. Artist developed tools have been integrated into the toolchains of large organisations, and a role for artists in the backend development of these organisations appears to be supported.

6.4.2 Existing Challenges. There is a clear challenge in establishing trust between artists and the organisations developing AI tools for artists. Past practices by AI organisations have damaged relationships with artists through large scale scraping of their artworks for training data. Resistance to the wider cultural adoption of AI is present, and exacerbated by concerns in changing economic landscapes. For a general support of art workers it is unclear where best to direct efforts; towards supporting traditional art practices facing automation, or towards emerging practices adopting new tools.

6.4.3 Areas for Future Research. How do public perceptions of AI-generated media and concerns about automated systems impacting employment affect the adoption and development of Creative AI tools?

Clarity on the public perception of AI generated media, and projections of the timeframes in which automated systems may start impacting employment, are needed in understanding what fears are misguided and what should be prepared for.

How can insights from past artistic projects within organisations inform the development of Creative AI tools, and what role can collaboration with artist communities play in this process?

Learning from artists past projects within organisations could help inform how best to bring artists' concerns into the development of Creative AI tools. Continued work within artist communities, both traditional and emerging, could be beneficial to researchers wanting to better understand the tools and support required.

6.5 Future Systems

6.5.1 Findings. The *Future Systems* theme looked at the reimagination of AI beyond current technical trends or possibilities. Ideas of co-creative partnerships with AI are being discussed and prototyped. Artists have shown how reimagining AI can allow for stronger cultural guidance, reinforcing a role for artistic thinking in the development of technology. There are endless varieties of what AI could be, and artists are developing ways of understanding AI beyond analytical intelligence through the imagination of distributed, living, multi-sensory, and cosmologically connected systems.

6.5.2 Existing Challenges. Working with speculative futures is difficult and runs into problems with current understandings of the limitations of technology. The current Co-Creative AI systems prototyped rely on the user suspending their disbelief and working within the fiction. This approach can confuse other discussed works in other themes that seek to demystify and explain current AI systems. Finding a balance between clarity of what is available today and what future we wish to imagine, without over-hyping or disguising the operation of AI requires focus.

6.5.3 Areas for Future Research. How can integrating current technologies into speculative AI futures improve our understanding of the technical contributions and interactions between human and non-human intelligences?

In a number of the works speculating on possible AI futures technology is often illustrative, rather than working with functional and reusable components. The tools and infrastructure that could be imagined for working with other intelligences are less developed. Finding ways to support the action of bringing present technologies into contract with more-than-human ideas of AI could help in clarifying the technical contributions that may otherwise lose their focus.

How does artistic research differ from other speculative design methods in exploring future AI systems, and what documentation and observational methods can best capture this work?

There are many possible approaches to speculating on future AI systems. Clarifying how artistic research differs from other speculative design methods, would help in communicating the benefits of this work. Alongside this, finding how best to document and observe this type of research can be further explored.

7 CONCLUSION

How have artists contributed to the development of artificial intelligence?

This review has highlighted the evolving relationship between artists and artificial intelligence, and how their contributions have informed the development of AI. Following this review it is evident that the field of study is in its

early and formative stages. Multiple authors have highlighted the lack of clarity on the impact the Arts have had on AI [22, 144]. This review is a first step in responding to this lack of clarity in understanding the roles that artists have in the technical and cultural development of AI.

The findings reveal how artistic contributions range from curation and analysis of datasets, to the development of tools and methods that can be used with AI, leading to the exploration of what AI could become. The study revealed the importance of novel forms of investigation that can further develop understanding of the field. Artists have been influential in developing AI and this work has been discussed in the academic research literature, but balancing artistic and research agendas remains problematic.

The findings from this review inform a wide range of future research trajectories. We propose directing attention toward two research trajectories: **Mapping Creative AI territories and entanglements** and **Developing speculative AI Devices through artistic practice**.

We will now discuss these further and draw research questions from them.

1) Mapping Creative AI territories and entanglements

There is a need to pursue further research into the roles of artists in this field. This is necessary because of the problematic relationship between cultural and technical forces in the development of artificial intelligence, and the divergent roles of academia and artistic practice. Understanding how artists integrate and navigate these technologies will clarify the evolving relationship. This has been explored at a formative stage in the diagrammatic representation of this survey in Figure 1.

Formulating this as a research question:

- *How can we map Creative AI territories and entanglements?*

2) Developing speculative AI Devices through artistic practice

It is through practice that we explore ways to support speculative and uncertain interventions in AI. This approach to investigating speculative AI devices generates the development of new methods, tools, and devices. Integrating current technologies into these speculations can help artistic work guide technological development. This work can yield both technical and cultural contributions, making it important to evaluate them on both fronts. These steps in the creative process can also be observed and analysed, where reflection on the process can be used in academic studies into the development of AI as guided by artistic practice.

We can guide these insights through the following questions:

- *How can we develop speculative AI Devices through artistic practice ?*
- *What processes are necessary in order to describe this work for others to be able to think about and use?*

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